

Description of Data

I chose to examine six different variables for this assignment. All variables were pulled from the GSS website. The variables and their descriptions are listed below.

Variable	Description
age	Age of respondent
educ	Highest year of school completed
paeduc	Highest year of school completed, father (or oldest same-sex parent)
maeduc	Highest year of of school completed, mother (or youngest same-sex parent)
Sex	Respondents sex
Race	Race of Respondent

With this data, the primary motivation is to analyze education trends and correlations. More specifically, this set of variables can help explore the correlation between the level of education of someone's parents and their own education. Education is an extremely informative variable and educational outcome can directly impact a person's income, occupation type, and their overall life opportunities. With education and parents' education, it is possible to analyze intergenerational patterns. Furthermore, it allows us to examine the impact of how someone's raised on their education. For example, if someone's parents both attended and completed college, they may be more likely to attend college than someone whose parents did not complete education beyond high school.

In addition to education and age data, sex and race are also included in this analysis. This deepens the value and relevance of this study, as it allows us to look at how someone's sex and

race can affect their educational outcomes. Historically, this has been a point of conversation in the United States. Looking at how race impacts education is crucial to identifying themes of system disparities. By combining these demographic variables with parental education patterns, the analysis can provide insights into how social context and possible inequalities relate to educational outcomes. These trends can help inform broader questions about educational attainment.

Visualizations and Tables

Table 1: Statistical Summary of Key Numerical Variables

	year	id_	educ_years	paeduc_years	maeduc_years
count	47844.000000	47844.000000	47844.000000	47844.000000	47844.000000
mean	1999.158599	1247.785260	13.826122	11.504138	11.624843
std	15.786098	900.070075	2.813038	3.842172	3.179408
min	1972.000000	1.000000	3.000000	3.000000	3.000000
25%	1986.000000	543.000000	12.000000	8.000000	10.000000
50%	1998.000000	1093.000000	14.000000	12.000000	12.000000
75%	2014.000000	1748.250000	16.000000	14.000000	13.000000
max	2024.000000	4509.000000	20.000000	20.000000	20.000000

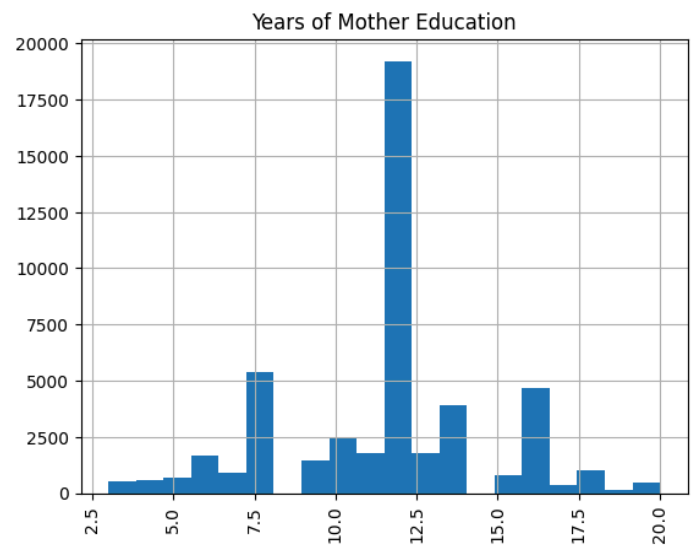
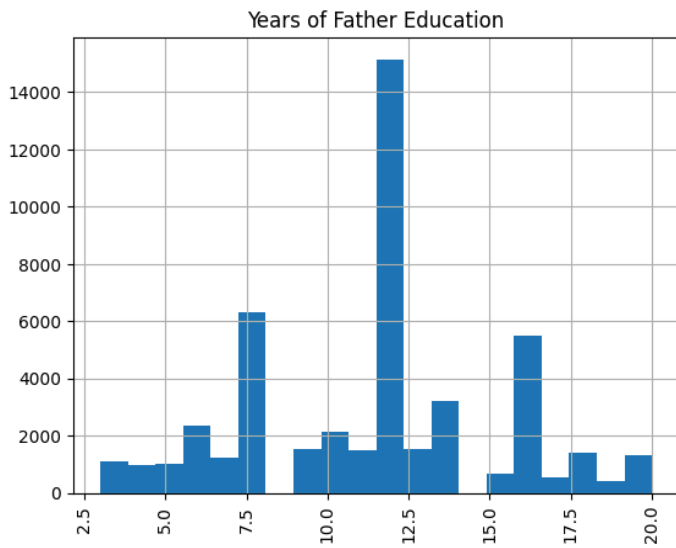
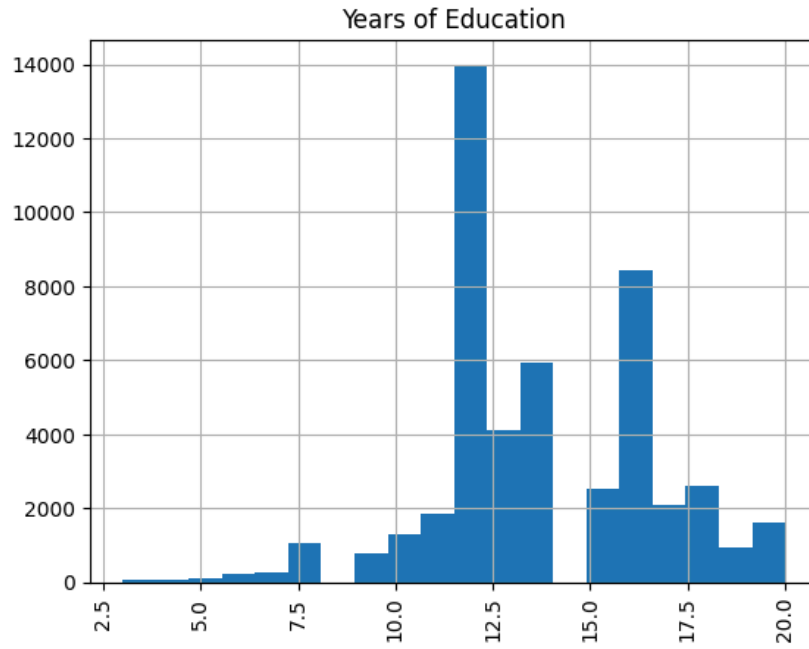
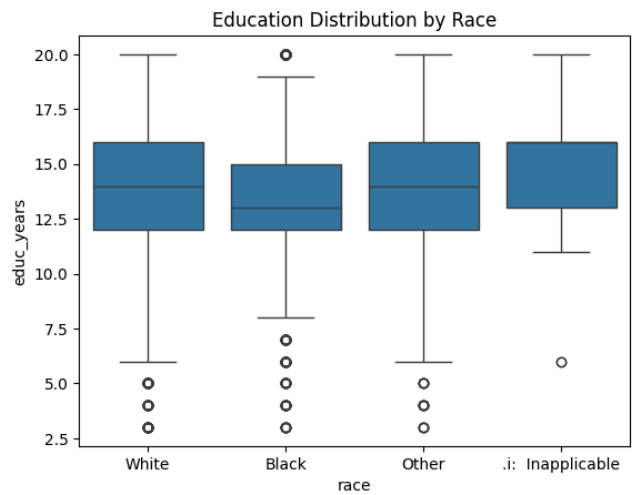
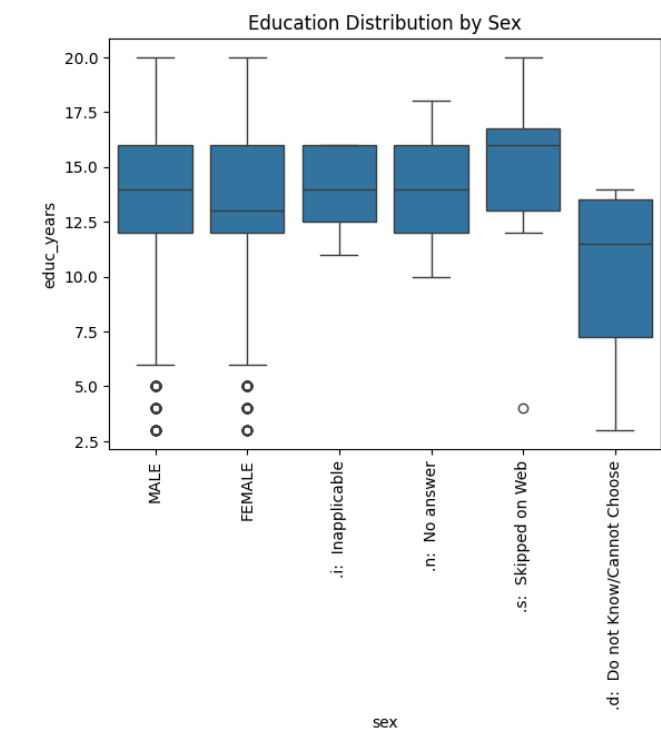
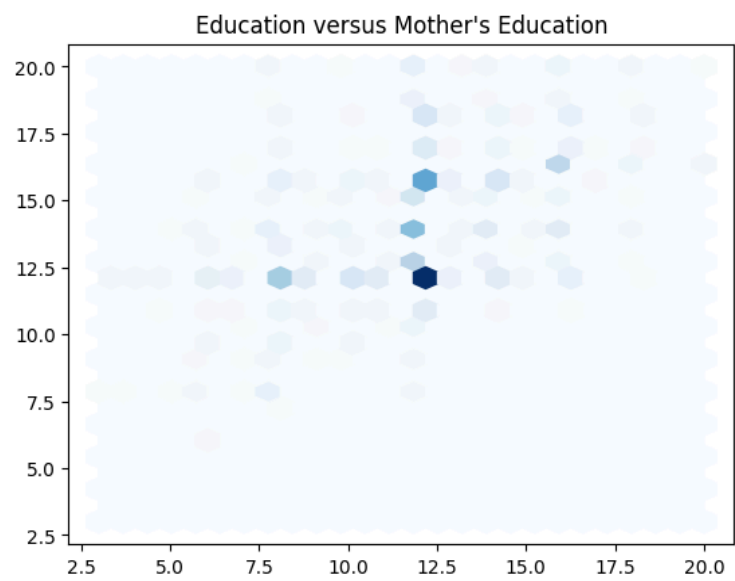
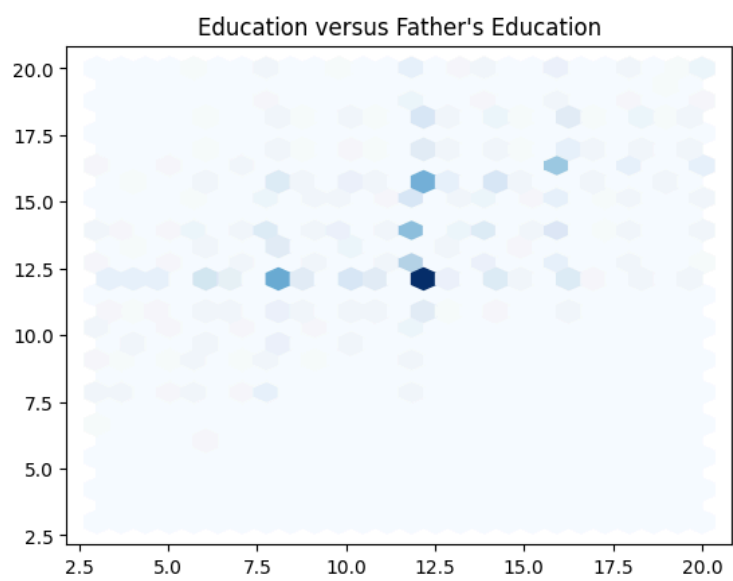


Table 2: Correlation between an Individual's Education and their Parents' Education

	educ_years	paeduc_years	maeduc_years
educ_years	1.000000	0.433136	0.419579
paeduc_years	0.433136	1.000000	0.636932
maeduc_years	0.419579	0.636932	1.000000



Results and Findings

Dataset Characteristics:

There are 47,844 observations in this dataset for the numerical variables. The minimum and maximum values all look normal and so do the mean and standard deviation of each variable. From first glance, the statistical summaries are already conveying information about the dataset. They show that the average education level is 13.8 years, meaning that the average person in the dataset received education past high school. Both parents have similar average education levels, with Father's average education year being 11.5 and Mother's average education year being 11.6. These values make sense, and it also makes sense that on average, respondents have completed more years of education than their parents. The histograms visualize these education variables. Looking at the histogram, most respondents reported something around 11-12 years of education, meaning they completed high school. Additionally, children have more activity in their college years than their parents.

Examining relationships:

The main relationship this analysis aimed to examine is the connection between a child's education level and their parents' education level. Examining the correlation between these variables shows that there is indeed a relationship. When someone's parents have received higher education, it is more likely for them to also pursue education beyond high school. Obviously, there are other factors at play that affect a child's education, but the level of education their parents received plays a crucial role. If someone's parents attended college, it is likely that they would have conversations about it at home, exposing their child to what higher education looks like.

A further relationship this analysis aimed to look at was the connection between race, sex, and education. This relationship is best looked at through the boxplots created using the dataset. The sex one shows that males have a higher median education level, but male and female results look relatively similar. Examining the boxplot comparing race and education reveals some interesting insights. White people report higher education levels, and black people have more outliers in the lower years, with 20 years of education also being an outlier for that group.

This data supports the understanding of education trends. With a more deeper analysis, education attainment could be split up into years, which could be used to analyze how much more education a person receives now than in the past. The year by year breakdown of data could also help to better understand changes in education by demographic, as females and non-white people had less access to higher educational opportunities in the past.